FEASIBILITY ASSESSMENT

Child Safety App for Android

What child-safety monitoring and restriction can actually do on an Android phone, and where the platform stops it.

Prepared September 2026 Based on a working build Test device: Google Pixel 4, Android 13

Summary

Most of what a parent expects from a child-safety app is substantially achievable on Android, and the greater part of it has been built and exercised on a device rather than estimated. The limits that remain come from Android platform behaviour, manufacturer variation and Play Store policy, not from the amount of development work, and a normal consumer app cannot engineer around them.

The right expectation to set with a family is a **strong, visible deterrent** rather than a sealed device. A child who is determined, patient and technically curious can find an edge. In ordinary day-to-day use, blocked apps close, explicit searches are shut down, and the parent can see that it happened.

9	7	3	8
Work reliably under the conditions tested	Work, with one named condition or blind spot	Deliver part of the outcome, not all of it	Platform and policy limits, documented in full

Nineteen capabilities were assessed, and all nineteen appear in the table below. Separately, eight limits that no normal consumer app can engineer around are listed further on, so nothing is left to be discovered during testing.

How to read the ratings

Fully feasible	Works reliably under the conditions tested, once the parent has given permission.
Feasible with conditions	Generally works as expected, subject to a named dependency or blind spot.
Partly feasible	Part of the outcome is achievable. The rest is limited by the platform.
Not achievable	Out of reach for a normal consumer app, for platform or policy reasons.

Capability by capability

CAPABILITY	RATING	WHAT THAT MEANS IN PRACTICE
RESTRICTING WHAT THE CHILD CAN OPEN		
Blocking chosen apps	Fully feasible	A switched-off app is closed when it is seen in the foreground, and the child is told which rule stopped it. Android gives no instant signal that the foreground app has changed, so there is a short detection and enforcement delay.
Daily time limits	Fully feasible	Limits can be set per app or per category. Once the day's allowance is spent, further attempts to open the app are closed, subject to the same short detection delay.
Bedtime and schedules	Fully feasible	Chosen categories are restricted inside a window the parent sets, and become available again on their own afterwards.

CAPABILITY	RATING	WHAT THAT MEANS IN PRACTICE
Blocking other VPN apps	Feasible with conditions	Recognised VPN apps are detected, closed as they open, and pushed off the single connection slot Android allows. A new or unrecognised VPN app, or another tunnelling method, may still succeed in connecting depending on the app, the Android version and the device.
Cloned and duplicate apps	Feasible with conditions	Recognised cloning tools and duplicate installs are detected and closed. A manufacturer's own clone space, such as Samsung Secure Folder, is not visible to a normal app, and a cloning tool that renames itself may not be recognised.
Shorts, Reels and TikTok feeds	Feasible with conditions	The short-video feed is recognised inside YouTube, Instagram, TikTok, Facebook and Snapchat and closed shortly after it opens. Recognition reads those apps' own screen layouts, so it needs ongoing maintenance as they redesign.
Status and Stories	Feasible with conditions	The same for WhatsApp Status, Instagram and Facebook Stories, Snapchat and Telegram, with the same maintenance need.
Resisting removal by the child	Partly feasible	Uninstalling can be made to require a deliberate extra step in Android Settings, and attempts to switch protection off are recorded and reported. On a consumer device it cannot be prevented outright.
FILTERING WHAT REACHES THE BROWSER		
Forcing SafeSearch everywhere	Fully feasible	While the filtering connection is running, Google, Bing, DuckDuckGo and YouTube are answered with each company's own filtered address, so SafeSearch applies in Chrome and cannot be switched off from within the search page.
Blocking sites and categories	Fully feasible	Whole categories, or individual sites the parent names, are refused before the page loads. An allow-list-only mode is available for younger children.
Closing restricted searches	Feasible with conditions	What is typed into a recognised browser's search bar is checked, and the browser is closed if the search is restricted. This catches the ordinary case; a child who deliberately words a search around the list can get through.
Ad and tracker removal	Feasible with conditions	Ads are refused across the device wherever they come from a recognised advertising source, while the filtering connection is running. Ads delivered from the same address as the content itself can only be removed inside our own browser and player.
MANAGING WHAT THE CHILD SEES		
Supervised YouTube viewing	Fully feasible	The built-in player runs with YouTube's Restricted Mode on, ads refused and Shorts unreachable. This is a surface the app controls, so behaviour here is predictable.
Covering unsafe images while browsing	Fully feasible	Inside the built-in browser, images are judged on the phone before the page shows them, and an image that fails arrives covered rather than intact. Judgement accuracy has a ceiling, described further on.

CAPABILITY	RATING	WHAT THAT MEANS IN PRACTICE
Covering unsafe video in other apps	Partly feasible	The screen is sampled several times a second and an overlay is placed over unsafe video while it plays. The app cannot alter or genuinely blur another app's pixels, so it covers rather than blurs. Screen monitoring also depends on Android's per-session capture consent, which a normal consumer app cannot hold silently or indefinitely.
AI chat use	Feasible with conditions	Time spent in recognised AI apps is measured against a daily budget, and what the child types is checked for concerning subjects without the text being stored. Detection is phrase-based, so unusual, coded or non-English wording is missed.
GIVING THE PARENT VISIBILITY		
Screen time and app usage	Fully feasible	Time for today, a per-app breakdown and a seven-day history, read from Android's own usage records. Where a figure genuinely cannot be read, the parent is told so rather than shown a zero.
Activity log and alerts	Fully feasible	Blocks, attempts and changes to protection are recorded, and serious events raise a notification. The log distinguishes what was actually stopped from what was only noticed.
Concerning-message signals	Partly feasible	Incoming messages that appear on the phone's notification shade can be checked for concerning content. Apps that hide message previews give nothing to read, and messages the child sends are never visible.

Location tracking and geofencing are not covered by this assessment. They raise no feasibility question on Android and can be added straightforwardly if the client wants them.

Limits that cannot be engineered around

These come from Android platform behaviour, from manufacturer differences, or from Google Play distribution policy, rather than from development decisions. They apply to a normal consumer app installed from the Play Store. Going beyond them means enrolling the phone as a managed device, which requires a factory reset and specialist setup and is a different product proposition.

> **Turning on Restricted Mode inside the official YouTube app.** Google exposes no interface that lets another app change that setting. The workable answer is to restrict the official app, which is enforced, and route the child to our supervised player.

> **Seeing inside a manufacturer's private space.** Samsung Secure Folder, Xiaomi Dual Apps and their equivalents are isolated by the operating system, and their contents are not visible to an ordinary app. The app reports that blind spot rather than declaring the phone clean.

> **Watching the screen permanently without asking again.** Screen monitoring runs on Android's screen-capture mechanism, which requires the user to consent to each session and cannot be maintained silently or indefinitely by a normal consumer app. Sessions end on restart and on certain system events; when one ends, a notification offers consent straight back.

> **Altering what another app draws on screen.** An app cannot modify or genuinely blur pixels rendered by another app's interface. The workable approach is an overlay placed on top, which hides rather than blurs: in the current build it leaves roughly one percent of the picture, enough to know an app is there and not enough to make out what is in it.

> **Filtering an app that uses encrypted lookups.** If the phone or a browser is pointed at a private encrypted lookup service, filtering cannot see that traffic, and a normal app cannot change that system setting. It is detected and reported to the parent instead.

> **Blocking an app with no visible flash.** Android does not notify an ordinary app when the foreground app changes, so a blocked app is briefly visible before it closes. The length of that gap depends on the Android version and the device, and it cannot be removed entirely.

> **Reading call logs and text messages.** Android does provide permissions for this, so the barrier is commercial rather than technical: Google Play policy restricts those permissions to apps whose core purpose is calling or messaging. A child-safety app is unlikely to qualify, and requesting them risks the listing being rejected or removed. It remains possible outside the Play Store, or with an approved policy exemption.

> **Making the app impossible to remove.** Full lock-down requires enrolling the phone as a managed device, which needs a factory reset and specialist setup. On an ordinary family phone, removal can be made awkward and clearly visible to the parent, but not impossible.

What setup asks of the parent

This is the single largest factor in whether the product works in a real household, and it is worth being direct about with the client.

- Most protections need a permission that Android will only grant through its own Settings screens, not through a simple yes-or-no prompt in the app. The app guides the parent through them in order of importance.
- Screen monitoring, which powers video covering, has to be allowed again for each new session. A notification offers it back so the parent is not left guessing.
- Battery optimisation has to be relaxed for this app, or the phone may pause it in the background and gaps appear in enforcement. Some manufacturers are more aggressive here than others.

- Nothing pretends to work without its permission. A protection that cannot run says so plainly and names what is missing, rather than showing a green tick.

How accurate the automatic judgement is

Two parts of the product judge content rather than simply matching a list, and both have a ceiling worth stating now.

- **Image and video judgement.** Android ships no ready-made explicit-content detector that runs on the phone, and the accurate ones require uploading the picture, which conflicts with the on-device privacy design. The current judgement uses a general-purpose picture-recognition model as a stand-in. It catches the obvious cases well and it is not a certified safety filter. A purpose-trained model would be needed before making a formal safety claim.
- **Restricted search wording.** The list of restricted terms is a floor rather than a complete map of everything a child might type. It is deliberately conservative, because a filter that closes the browser during homework is a filter the parent switches off within a week.

Privacy position

> The current build is designed to judge everything on the phone. It has no server component, and no network call in the implementation carries an observed message, screen image, search or web address off the device.
>
> What is kept is the verdict rather than the content that produced it: how long, which category, how risky, which site. Message text, anything typed into an AI, screen images and full web addresses are examined in memory and discarded. This describes the architecture as implemented; an independent privacy review would be the normal step before it is offered as a formal guarantee.

What was exercised on a real device

The behaviour below was observed directly on a Pixel 4 running Android 13. Ratings elsewhere in this document that are not listed here rest on the implemented behaviour and on Android's documented platform rules rather than on a device test of every case, and no single device can represent every manufacturer's build.

- A blocked app closed on each launch attempted during testing, with the block screen naming the rule that stopped it.
- Searching for a restricted term in Chrome closed Chrome and alerted the parent. An ordinary homework search in the same browser was left alone.
- With filtering on, Google and YouTube resolved to their forced-safe addresses while ordinary sites were untouched, and encrypted lookup services were refused.
- A gambling site was refused before the page was fetched, and the category was named to the child.
- A real third-party VPN was detected, the parent was alerted, and the reason it could not be removed was stated precisely.
- Over an image search, no photograph on the screen remained recognisable behind the cover, and the cover lifted by itself once the screen was clean again.
- With usage permission withheld on purpose, the dashboard reported that screen time was unavailable and named the missing permission, instead of showing zero.

Conclusion

The product is substantially feasible on Android and is largely built. Of the nineteen capabilities assessed, sixteen deliver their intended outcome either reliably or with one named condition, and the remaining three deliver a useful part of it. The eight limits set out above are documented rather than suspected, which is what makes the rest of the estimate dependable.

What should be promised to a family is a phone where the ordinary routes to unsuitable content are closed, where time limits hold, and where the parent can see what happened. What should not be promised is a device a determined teenager cannot outwit. That distinction follows from how Android works for consumer applications, and stating it plainly at the outset is what keeps a parent's trust once they start using the product.